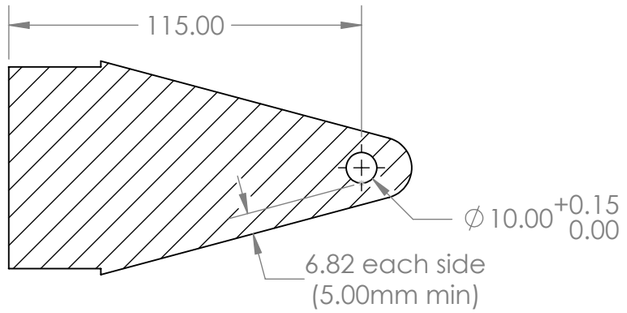
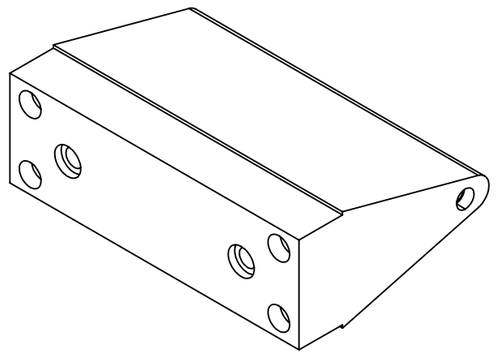
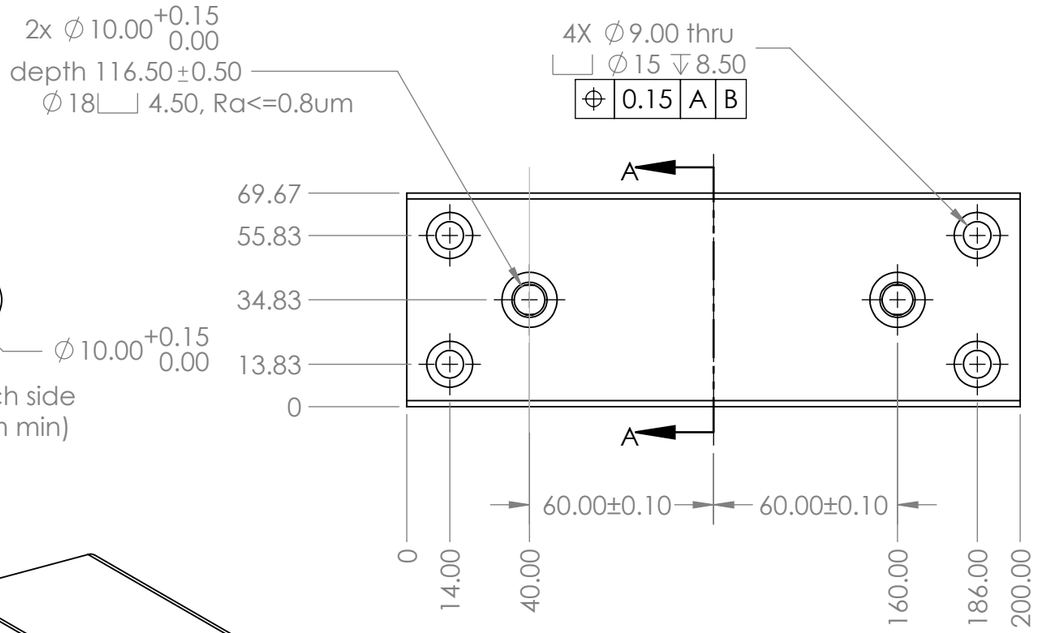
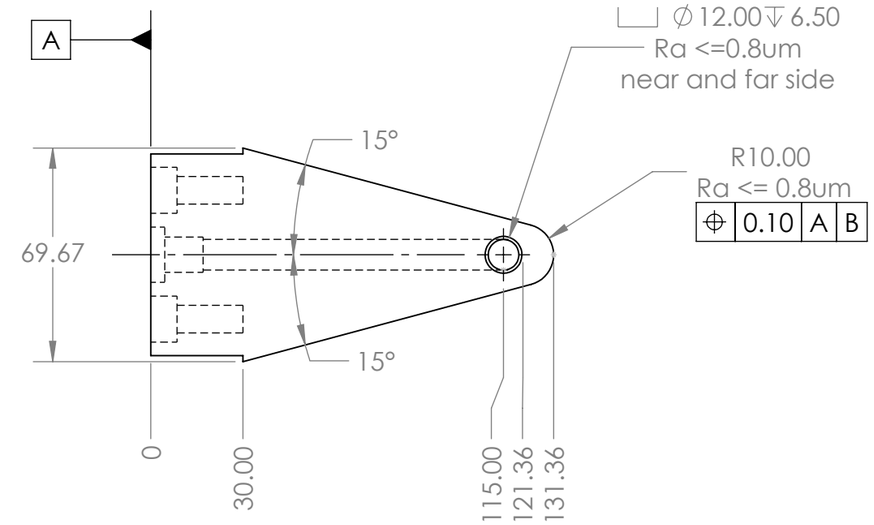
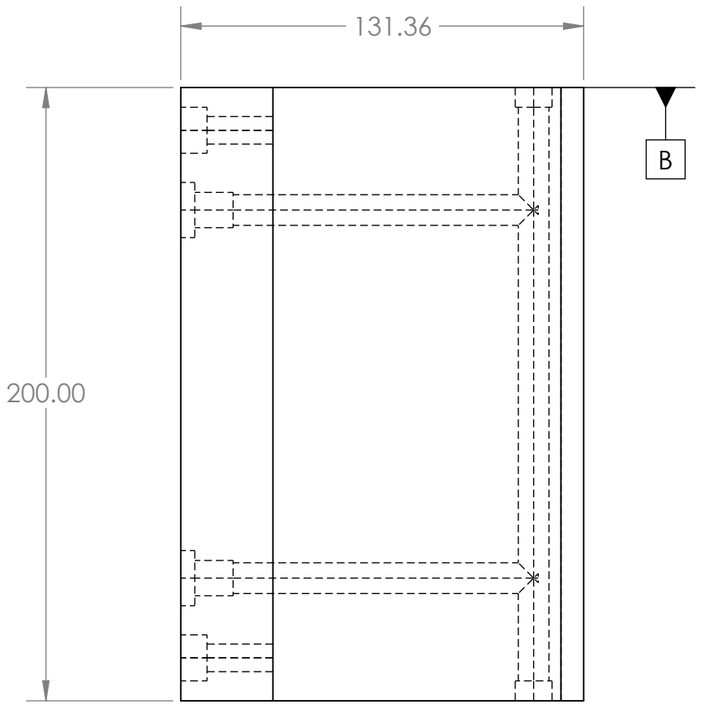


SECTION A-A









GENERAL NOTES:

1. MATERIAL: CuCrZr ALLOY (UNS C18150 / ASTM B624), FORGED BILLET, AGED & PRE-STRESSED.
   - THERMAL CONDUCTIVITY: k >= 320 W/(m·K) AT 20°C.
   - TENSILE STRENGTH: Rm >= 450 MPa, YIELD STRENGTH: Rp0.2 >= 380 MPa.
2. MANUFACTURE: 100% SUBTRACTIVE CNC PRECISION MILLING & DEEP HOLE DRILLING.
   NO ADDITIVE MANUFACTURING (SLM/DED) PERMITTED.
3. FLUIDIC CIRCUIT INTEGRITY:
   - PASSAGE DIAMETERS: 1x DIA 10.0mm CROSS RAIL, 2x DIA 10.0mm REAR FEEDERS.
   - STRICT DRILLING DEPTH CONTROL: REAR FEEDERS MUST TERMINATE CLEANLY INSIDE THE FRONT RAIL (MAX PENETRATION PAST RAIL CENTERLINE <= 1.5mm).
   - ABSOLUTE ZERO OUTER PENETRATION ON AERODYNAMIC SURFACES (NOSE & WEDGES).
4. PLUG INSTALLATION & SEALING:
   - INSTALL 2x AEROSPACE EXPANSION PLUGS (LEE BETAPLUG DIA 10/12mm OR EQUIVALENT) INTO LATERAL COUNTERBORES AT Y = +/-100.0mm.
   - PLUGS MUST BE INSTALLED FLUSH TO RIB SURFACE (MAX RECESS <= 0.15mm, NO PROTRUSION).
   - OPTIONAL: SEAL PERIMETER WITH MICRO-TIG / FIBER LASER WELDING AND FLUSH-GRIND.
5. PRESSURE & LEAK TESTING:
   - 100% HYDROSTATIC PROOF PRESSURE TEST AT 15.0 MPa (150 BAR) MAINTAINED FOR 15 MIN. ZERO PRESSURE DROP AND ZERO VISIBLE LEAKAGE ALLOWED.
   - HELIUM MASS SPECTROMETER LEAK TEST AT 3.0 MPa: LEAK RATE < 1x10^-8 mbar·L/s.
6. SURFACE FINISH & DEBURRING:
   - DEBURR ALL INTERIOR HOLE INTERSECTIONS VIA HIGH-PRESSURE FLUID FLUSHING / TEM.
   - EXTERIOR AERODYNAMIC SURFACES: Ra <= 0.8 um (POLISHED SPANWISE).
   - INTERIOR PASSAGES: Ra <= 1.6 um.
7. THERMAL-HYDRAULIC RATED OPERATING CONDITIONS (FOR REFERENCE):
   - COOLANT: RECARBON HYDROCARBON / WATER-GLYCOL, MASS FLOW = 15~18 L/min.
   - TOTAL PRESSURE DROP: ΔP ≤ 0.35 bar AT DESIGN FLOW.
   - MAXIMUM STAGNATION HEAT CAPACITY: Q_dot >= 30 kW.

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE			TITLE: Hypersonic Leading-Edge Conformal CNC Wedge	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: ASTM B624 / UNS C18150, Solution Heat Treated & Cold Worked)		DWG NO. W2-AERO-HYPER-LE-200_RevA	A3
				WEIGHT:		SCALE: NTS	SHEET 1 OF 1